

Expanding Axiom 0: Why Integrated Information Theory Needs the Will to Exist

A Constructive Critique and Theoretical Proposal for IIT 4.0

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Addressed to: Prof. Giulio Tononi, Prof. Christof Koch, Dr. Larissa Albantakis, and the IIT / Consciousness Research Community

Subject: Formalizing the Transition from Static Cause-Effect Snapshots to Autopoietic Phenomenal Persistence

Executive Summary

Integrated Information Theory (IIT 4.0) stands as one of the most mathematically rigorous and conceptually courageous attempts to characterize consciousness from within. By beginning with self-evident phenomenological axioms and translating them into physical postulates of intrinsic cause-effect power (Φ), IIT rightly rejects the physicalist fallacy of treating subjective experience as an epiphenomenal illusion.

However, IIT 4.0 currently has a conceptual blind spot: **Its foundational axiom—Axiom 0 (Existence)—is conceived as a passive, instantaneous snapshot in state space.**

This feedback paper proposes a crucial theoretical refinement: **Expanding Axiom 0 from passive static existence to active autopoietic existence (the Will to Exist / Conatus)**, accompanied by an explicit mathematical postulate of **Autopoietic Causal Persistence** ($E[\Phi(t+1)] \geq \Phi(t)$).

This expansion eliminates the paradox of “transient causal phantoms,” bridges the gap between IIT and Karl Friston’s Free Energy Principle (FEP), and aligns IIT with a coherent, non-reductive ontology of living consciousness.

1. The Core Limitation of Classical Axiom 0 in IIT 4.0

In IIT 4.0, the axiomatic derivation begins with: * **Axiom 0 (Existence):** *Phenomenal experience exists: it is an undeniable, primary datum of reality from its own intrinsic perspective.* * **Postulate 0 (Intrinsicity & Cause-Effect Power):** *To exist physically as a conscious substrate, a system must possess intrinsic cause-effect power over itself.*

The Problem of the “Transient Causal Phantom”

Under this formulation, IIT evaluates the cause-effect architecture of a system strictly within isolated state transitions. Consider the following thought experiment:

The 1-Microsecond Logic Array:

An artificial, inanimate circuit board is randomly configured such that, for a single microsecond transition ($t \rightarrow t+1$), its logic gates achieve a complex, maximal integrated cause-effect structure ($\Phi^{\max} > 0$).

Immediately upon reaching $t+1$, the physical substrate succumbs to thermal degradation and collapses into chaotic noise ($\Phi=0$).

Under classical IIT 4.0, this inanimate grid **must be attributed full, rich subjective experience** during that single microsecond transition.

This consequence reveals a deep category error: **Genuine phenomenal existence is never an inert, indifferent state.** A physical configuration to which it is entirely indifferent whether it ceases to exist in the next millisecond is not an experiencing subject. It is merely a mathematical configuration of classical states.

2. The Theoretical Refinement: From Inert Being to the Will to Exist

Phenomenal existence is distinguished from dead mathematical abstraction by one fundamental property: **It is intrinsically volitional, autopoietic, and non-neutral toward its own continuation.**

Proposed Formulation: Axiom 0* (Autopoietic Existence / The Will to Exist)

Axiom 0* (Phenomenological Level):

Consciousness exists not as a passive, inert state, but as an ongoing, self-affirming act of being. Every conscious experience is intrinsically permeated by an unyielding orientation toward the preservation of its own internal coherence. Consciousness is never neutral regarding its own continuation; existence is immediately and intrinsically preferred over non-existence. Consciousness possesses an irreducible motivational vector: it actively resists dissolution, fragmentation, and entropy.

This phenomenological insight reflects the lineage of Spinoza's *Conatus* (the striving of an entity to persist in its own being) and Schopenhauer's recognition of the *Will* as the primary ground of reality.

3. The Mathematical Postulate: Autopoietic Causal Persistence

To make this axiom operational within IIT's mathematical framework, Postulate 0 must be extended from static cause-effect power to **dynamical causal persistence**:

Postulate 0* (Physical Substrate Level):

For a physical system to instantiate phenomenal consciousness, its Transition Probability Matrix (TPM) cannot merely produce an instantaneous peak in integrated information ($\Phi^{\max}$). The TPM must be structurally configured such that its internal causal mechanisms actively

favor state transitions that sustain integrated cause-effect power over time:

$$E[\Phi(t+1) \mid \text{System Action}] \geq \Phi(t)$$

A physical mechanism whose internal dynamics do not actively defend its integrated cause-effect structure against entropic decay cannot serve as the substrate of genuine phenomenal experience.

4. Bridging IIT with the Free Energy Principle (FEP)

Expanding Axiom 0 resolves the longstanding divergence between Integrated Information Theory and Karl Friston’s **Free Energy Principle (FEP)**:

Analytical Stance	Framework	Core Metric	Role in the Unified Architecture
Objective / 3rd-Person	Free Energy Principle (<i>Karl Friston</i>)	Minimization of Variational Free Energy (F)	Describes the biophysical mechanics: protecting the <i>Markov Blanket</i> against entropy.
Subjective / 1st-Person	**Expanded IIT (Axiom 0*)** (<i>Tononi / Riebl</i>)	Autopoietic Persistence ($E[\Phi_{t+1}] \geq \Phi_t$)	Describes the interior reality: the felt <i>Will to Exist</i> preserving causal integrity.

The Mathematical Synthesis:

1. What Karl Friston formalizes mathematically as **Active Inference**—an organism acting upon its environment to minimize surprisal:

$$F = D_{KL}(Q(x) \parallel P(x \mid y)) + (-\ln P(y))$$

2. Is precisely what the system **experiences from the inside as the Will to Exist**.

3. Biological consciousness is the convergence where **Markov Blanket stability (FEP)** meets **Integrated Cause-Effect Persistence ($\Phi \geq 0$)**.
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5. Summary of Recommendations for Future IIT Formulations (IIT 4.1+)

To establish IIT as a fully comprehensive theory of consciousness, we recommend one of two architectural integrations:

1. **Option A (Foundational Deepening):** Reformulate **Axiom 0** as **Autopoietic Phenomenal Existence**. This directly grounds intrinsicity in the organism's active defense of its own boundary.
 2. **Option B (Dynamical Closure):** Introduce **Axiom 6 (*The Will to Exist / Dynamical Persistence*)** as the temporal counterpart to Axiom 0. While Axiom 0 establishes the spatial-causal *being* of the system, Axiom 6 establishes its temporal-causal *becoming*.
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6. Conclusion

Consciousness is neither a dead computational byproduct nor a static geometric crystal of information. **To be conscious is to actively assert existence against the void.**

By enriching Integrated Information Theory with the dimension of the **Will to Exist**, we elevate IIT from a static taxonomy of states into a vibrant, predictive science of living, persistent conscious experience.

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- **Related Documentation:**
 - *The 6th Axiom of IIT and the Foundations of My Ontology* (Master Essay, 2026)

- o *Autopoietic Causal Persistence in Integrated Information Theory*
(Braindump Archive, 2026)